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Fault identification method based on unified inverse-time characteristic equation for distribution network☆

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ABSTRACT

Traditional inverse-time overcurrent (ITOC) relay adapts itself to the severity of the fault and can be used for fault identification in the main/backup relays. A fault identification method based on a unified inverse-time characteristic equation is proposed to aim at the problems of large setting workload and easy misoperation of inverse-time overcurrent relay after distributed generation access. It does not require a step-by-step setting and communication, and can realize adaptive coordination of multilevel relays. Firstly, the distribution law of positive sequence fault component voltage and the influence of distributed generation on its distribution characteristics are analyzed. Secondly, the setting principles and problems in implementing a unified equation for the multilevel coordinated relays are analyzed for the standard ITOC characteristic equation. The functional form of the user-defined inverse-time characteristic equation is derived from the ITOC equation and the setting principles considering the influence of fault conditions are given. The expression of the unified inverse-time characteristic equation is given, and the action sequencing of the adaptive fault position is obtained from the calculated tripping time. Finally, the simulation results validate the effectiveness of the proposed method.

1. Introduction

Protection devices in series are time-coordinated, such as circuit breakers and fuses [1,2]. With the increasing complexity of power grid structures, the setting workload of traditional non-unit protection based on time delay coordination is very large [3,4]. This flexible and changeable operation mode sometimes leads to failure and misoperation of relays [5]. Currently, most wide-area backup protection devices require communication, and the reliability and expense of communication technology limit their engineering application [6,7]. Thus, there is an urgent need to study fault identification methods independent of communication and tripping time to realize adaptive coordination of multilevel relays.

ITOC relays automatically adjust the tripping time according to the current measurement. They are widely used in medium- and low-voltage distribution networks due to their simplicity and economy [8,9]. Inverse-time characteristic parameters are set in the fixed network topology so that the devices operate in a specified sequence for distribution network coordination of multilevel relays [10]. Distributed generations (DGs) access does not change the topology, but different locations and numbers of DGs have different effects on the upstream and downstream protective devices. Timing coordination between relays is affected due to the additional fault current of DGs. The fuse cannot distinguish the direction, and DG access leads to the loss of selectivity in the coordination between it and other devices [1,11]. For these cases, selectivity may have to be studied on a case-by-case basis [12]. In modern and future distribution networks, spurs are connected to the main feeder through spur sectionalizes rather than fuses [13].

Many methods have been proposed to improve ITOC relays coordination. In [14,15], the fault voltage is directly introduced into ITOC. By combining the voltage with ITOC, an inverse-time low-impedance protection scheme was proposed in [16]. The introduction of voltage makes the relays coordinated and suitable for DG access. Nevertheless, it does not solve the problem of a large setting workload and may further increase the setting workload. In [17–19], the coordination problem of main/backup relays was transformed into an optimization problem,

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such as genetic algorithms [17], metaheuristic optimization approaches [18], mixed-integer linear programming [19], etc. They can reduce the setting workload and improve selectivity and rapidity, but until now, there has been no unified expression for objective functions.

Aiming at the problem that it is difficult to set the relays setting value after DG access, the protection schemes are adjusted in real-time to respond to the changing system conditions and achieve the adaptive coordination of relays [20]. Adaptive overcurrent relay updates setting online, which has the problems of insufficient response speed or relying on communication [20,21]. In summary, it is necessary to study the fault identification method without step-by-step setting and adaptive fault position.

Digital overcurrent relay (DOR) has the advantage of programmability, so the user-defined inverse-time equation is gradually applied [22–24]. In [23], the relay characteristic constant and the inverse-time type constant are also optimized as variables. The logarithmic function is proposed in [24], and the inverse-time coefficients of all relays are equal. In addition, the fault component voltage appears only in the fault state, and there is no dead-zone problem. Its distribution characteristic is opposite to the fault voltage [25]. Consequently, by combining the nonstandard characteristic equation with the fault component voltage, studying the fault identification method based on a unified inverse-time characteristic equation of all relays is meaningful.

A fault identification method based on adaptive coordination of multilevel relays is proposed. It utilizes the natural distribution characteristics of the positive sequence fault component voltage (PSFCV) and adopt a unified pickup value for all relays. Since the DG penetration needs to be known in advance when setting the time dial setting (*TDS*) of the standard ITOC characteristic equation, a user-defined inverse-time characteristic functional form and setting principle are given that does not require step-by-step setting. The inverse-time tripping time for each relay is adjusted to the current fault condition. The proposed method automatically identifies faults at any point and obtains the action sequencing of adaptive fault position, greatly reducing the setting workload.

The paper is organized as follows. Section II analyzes the distribution characteristics of PSFCVs and the influence of DG access on the PSFCV amplitude of all relays. Section III describes the inverse-time characteristic equation based on the PSFCV and parameter selection principle. Section IV gives the fault identification scheme of multilevel coordination relays. Section V presents simulations of a typical radial distribution network and a modified IEEE30-bus system to verify the effectiveness under different fault conditions and different DG penetrations. Finally, conclusions are drawn in Section VI.

2. Distribution characteristics of PSFCVs with a high proportion of DG access

2.1. Fault output characteristic of DGs

Traditional analysis method is not applicable to inverter interfaced distributed generations (IIDGs), which are connected to the grid through inverters [26]. Its response to fault depends on the control strategy and is highly non-linear. Therefore, this paper mainly discusses IIDGs.

IIDGs must have low voltage ride-through (LVRT) capability according to the rules of the German grid code [27]. They are not disconnected during a voltage drop down to 0 % of the rated voltage U_N, within the duration of $\leq$ 150 ms [28]. To improve the output characteristics, IIDGs generally adopt a positive-sequence control strategy [29]. Therefore, the relationship between the voltage sag γ and the output current I_g is shown in Fig. 1.

where $\gamma = U_f/U_N$ and U_f is the grid voltage at the grid connection point. I_{max} is the maximum fault current of the DG output, and I_{DG} is the pre-fault current generated by the IIDG.

It can be seen from Fig. 1 that the output currents I_g under different voltage sags are:

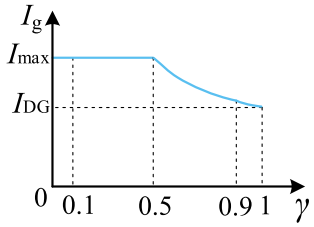


Fig. 1. Relationship between the voltage sag and output current.

$$I_g = \begin{cases} I_{max} & 0 < \gamma \leq 0.5 \\ \sqrt{I_d^2 + I_q^2} & 0.5 < \gamma \leq 0.9 \\ P_{ref}/U_{df} & 0.9 < \gamma \leq 1 \end{cases} \quad (1)$$

where I_d and I_q are the active and reactive output currents of the IIDG after the fault, P_{ref} is the active reference power, and U_{df} is the d-axis grid voltage at the grid connection point.

Thus, IIDGs provide a short-circuit current to the system. Its fault output characteristic can be equivalent to the voltage controlled current source in steady-state fault [30], as shown in Fig. 2.

where Z_{DG} is the equivalent impedance of DG, $\Delta\dot{I}_{DG}$ is the fault component current provided by DG without considering the current limiting strategy, and $\Delta\dot{I}_g$ is the fault component current provided by DG to the power grid.

2.2. Distribution characteristics of PSFCVs

Fig. 3 shows the radial network with DG connected to buses B and C. $R_1 \sim R_3$ are the relay numbers. m and n are the numbers of DG_1 and DG_2, respectively. The DG capacity is changed by changing m and n. f shows that the fault occurs at any position of L3.

1) Three-phase (LLL) fault: When a LLL metallic fault occurs at point f, the positive sequence fault component (PSFC) additional network is shown in Fig. 4.

where Z_{s1} and $\Delta\dot{I}_{s1}$ are equivalent positive sequence impedance and PSFC current of the system side, respectively. $m\Delta\dot{I}_{g11}$ and $n\Delta\dot{I}_{g12}$ are PSFC currents provided by mDG$_1$ and nDG$_2$. z_1 is per unit length of positive sequence impedance. $L_1 \sim L_3$ are the lengths of L1 ~ L3, and x is the distance between point f and bus C. $Z_{L11} \sim Z_{L13}$ and $\Delta\dot{I}_{L11} \sim \Delta\dot{I}_{L13}$ are the equivalent positive sequence impedance and PSFC currents of Ld1 ~ Ld3. $\Delta\dot{I}_{11} \sim \Delta\dot{I}_{13}$ and $\Delta\dot{U}_{11} \sim \Delta\dot{U}_{13}$ are the PSFC currents and PSFCVs of $R_1 \sim R_3$. Z_{feq1} is the equivalent positive sequence impedance of the line and load on the right side of point f. $-\dot{U}_{x|0|}$ is the pre-fault positive sequence voltage of point f. The specified current direction is from bus to line.

The PSFC currents flowing through S, DG_1, DG_2 and loads in the fault additional network are from line to bus, as shown by the solid red line in Fig. 4. Therefore, the actual direction of PSFC currents of all relays is from bus to line.

In Fig. 4, the PSFCVs of all relays can be expressed as:

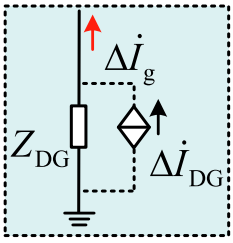


Fig. 2. IIDG's equivalent circuit in the fault component network.

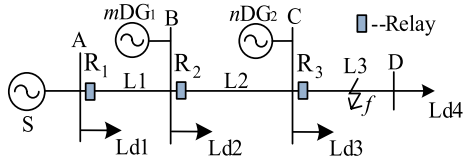


Fig. 3. Radial network with DG.

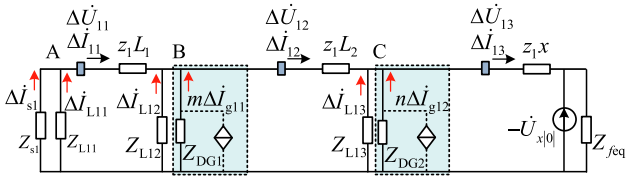


Fig. 4. The PSFC additional network.

$$\begin{cases} \Delta\dot{U}_{13} = -\dot{U}_{x|0|} + z_1 x \Delta\dot{I}_{13} \\ \Delta\dot{U}_{12} = \Delta\dot{U}_{13} + z_1 L_2 \Delta\dot{I}_{12} \\ \Delta\dot{U}_{11} = \Delta\dot{U}_{12} + z_1 L_1 \Delta\dot{I}_{11} \end{cases} \tag{2}$$

Affected by the voltage drop along the line, there is the following relationship:

$$|\Delta U_{13}| \rangle |\Delta U_{12}| > |\Delta U_{11}| \tag{3}$$

According to (2), the amplitudes of the PSFCVs are inversely proportional to the fault position. Fig. 5 shows the PSFCV distribution curves of each relay under different fault positions.

The closer a relay is to the fault point, the greater its PSFCV value. Its natural distribution characteristics provide a physical basis for obtaining a unified inverse-time characteristic equation without a step-by-step setting.

2) Phase-to-phase (LL) or phase-to-phase-to-ground (LLG) fault: When a LL metallic fault occurs at point *f*, the complex sequence network of fault component additional network is shown in Fig. 6.

where Z_{wAG} is the equivalent impedance of *w*-sequence at the back side of bus A (*w* is 1 or 2), and $\Delta\dot{U}_{wf}^{(2)}$ and $\Delta\dot{I}_{wf}^{(2)}$ are the equivalent fault component voltage and current of *w*-sequence at point *f* respectively. Superscript (2) indicates LL fault.

In Fig. 6, the PSFCVs of $R_1 \sim R_3$ can be expressed as:

$$\begin{cases} \Delta\dot{U}_{13}^{(2)} = \Delta\dot{U}_{1f}^{(2)} + z_1 x \Delta\dot{I}_{13}^{(2)} \\ \Delta\dot{U}_{12}^{(2)} = \Delta\dot{U}_{13}^{(2)} + z_1 L_2 \Delta\dot{I}_{12}^{(2)} \\ \Delta\dot{U}_{11}^{(2)} = \Delta\dot{U}_{12}^{(2)} + z_1 L_1 \Delta\dot{I}_{11}^{(2)} \end{cases} \tag{4}$$

Therefore, the PSFCV during a LL fault still satisfies the distribution characteristics of (3). However, it is known from the boundary condition that $\Delta\dot{U}_{1f}^{(2)} = -\dot{U}_{x|0|}/2$, the PSFCV amplitude of each relay is much smaller than that of the corresponding LLL fault.

When a LLG metallic fault occurs at point *f*, the complex sequence

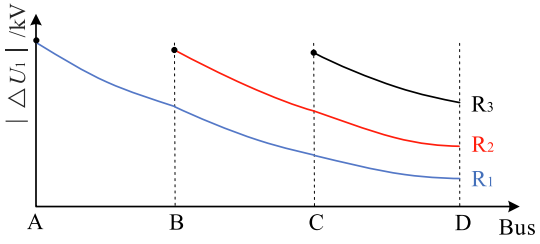


Fig. 5. PSFCV distribution curves of each relay under different fault positions.

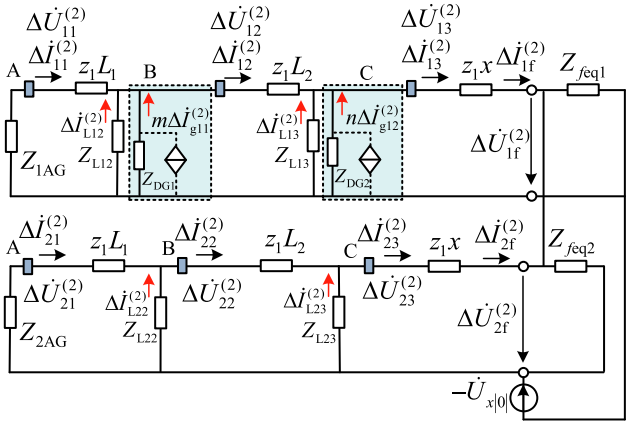


Fig. 6. Complex sequence network of fault component additional network.

network is the parallel connection of positive, negative and zero sequence networks. The PSFCV for each relay still has the form of (4). The boundary condition of point *f* is:

$$\begin{cases} \Delta\dot{I}_{1f}^{(1,1)} = \dfrac{-\dot{U}_{x|0|}}{Z_{\sum 1} + \dfrac{Z_{\sum 2} Z_{\sum 0}}{Z_{\sum 2} + Z_{\sum 0}}}, \ \Delta\dot{U}_{1f}^{(1,1)} = Z_{\sum 1} \Delta\dot{I}_{1f}^{(1,1)} \\ \Delta\dot{I}_{2f}^{(1,1)} = \dfrac{Z_{\sum 0}}{Z_{\sum 2} + Z_{\sum 0}} \Delta\dot{I}_{1f}^{(1,1)}, \ \Delta\dot{U}_{2f}^{(1,1)} = Z_{\sum 2} \Delta\dot{I}_{2f}^{(1,1)} \end{cases} \tag{5}$$

where $Z_{\sum 1}$, $Z_{\sum 2}$, and $Z_{\sum 0}$ are network equivalent impedances of positive, negative, and zero sequences, respectively. Superscript (1,1) indicates LLG fault.

When a LL or LLG metallic fault occurs at point *f*, it can be seen from the complex sequence network that the sum of the amplitudes of PSFCV and negative sequence fault component voltage at point *f* is equal to the PSFCV value for a LLL fault [31]. (6) can be deduced from the boundary conditions.

$$\begin{cases} \left|\Delta U_{1f}^{(2)}\right| + \left|\Delta U_{2f}^{(2)}\right| = U_{x|0|} \\ \left|\Delta U_{1f}^{(1,1)}\right| + \left|\Delta U_{2f}^{(1,1)}\right| = U_{x|0|} \end{cases} \tag{6}$$

Under different fault types, the PSFC networks on the left side of point *f* are similar, so the fault type coefficient k_{dj} is introduced to eliminate the influence of fault type on PSFCV. The value of k_{dj} is:

$$k_{dj} = \frac{|\Delta U_{1j}| + |\Delta U_{2j}|}{|\Delta U_{1j}|} \tag{7}$$

When a metallic short-circuit fault occurs, k_{dj} completely eliminates the effect of the fault types on the PSFCV. When a fault occurs via the fault resistence R_g, the voltage drop across R_g results in different PSFCV under different fault types. However, the introduction of k_{dj} can reduce the effect of fault types on the PSFCV voltage. Therefore, the following PSFCVs are revised by k_{dj}.

2.3. Influence of DG access on PSFCVs distribution

This paper uses the full cycle subtraction method to obtain the fault component voltages. Then, the fast Fourier transform technique is used to calculate the PSFCVs. Therefore, PSFCVs are related to both pre-fault voltages and fault positions.

It is known from [32,33] that DG access will support the grid voltage, which is closely related to DG capacity and access location. The larger the DG capacity is, the stronger the voltage support and the higher the pre-fault voltage level. It can be inferred from Fig. 4 that when the DG capacities increase, the PSFCVs of R_2 and R_3 still satisfy the relationship

in (2).

To realize that all relays adopt a unified characteristic equation, the adjacent relays at the most serious fault position should meet the selectivity requirements. The PSFCV distribution laws of R_3 and R_2 are qualitatively analyzed when the metallic fault occurs at the remote-end of L3. At this time, x is close to 0, and the PSFCVs of R_3 and R_2 are:

$$\begin{cases} \Delta\dot{U}_{13} = -\dot{U}_{x|0|} \\ \Delta\dot{U}_{12} = -\dot{U}_{x|0|} + z_1 L_2 \left(\Delta\dot{I}_{11} + \Delta\dot{I}_{L12} + m\Delta\dot{I}_{g11} \right) \end{cases} \tag{8}$$

The PSFCV difference $\triangle U_{R23}$ between R_3 and R_2 can be expressed as:

$$\Delta\dot{U}_{R23} = z_1 L_2 \Delta\dot{I}_{12} = z_1 L_2 \left(\Delta\dot{I}_{11} + \Delta\dot{I}_{L12} + m\Delta\dot{I}_{g11} \right) \tag{9}$$

Fig. 7 shows the effect of DG access on PSFCVs when the fault occurs at the remote-end of L3.

It is known from Fig. 7 (a) that the supporting effect of DGs on the bus voltage increases pre-fault voltage $-\dot{U}_{x|0|}$, resulting in the increase in $|\triangle U_{13}|$, which increases with the increase in m and n. As seen from (8) and Fig. 7 (a), the increase in m increases$z_1 L_2(\Delta\dot{I}_{11} + \Delta\dot{I}_{L12} + m\Delta\dot{I}_{g11})$, resulting in the decrease in $|\triangle U_{12}|$. The increase in n supports the pre-fault voltage and increases $\triangle U_{12}$. Therefore, the combined action of $m\Delta\dot{I}_{g11}$ and $n\Delta\dot{I}_{g12}$ leads to a change in the PSFCV value. However, a high proportion of DG access does not change the PSFCVs distribution characteristics of R_2 and R_3.

Fig. 7 (b) shows that $|\triangle U_{R23}|$ is mainly related to the number of m; that is, the PSFCV difference between adjacent upstream and downstream relays mainly depends on the supporting effect of the upstream line. Furthermore, a, b, c and d correspond to the value of $|\triangle U_{R23}|$ in four scenarios, indicating that DG_1 and DG_2 exit or output at maximum capacity, respectively.

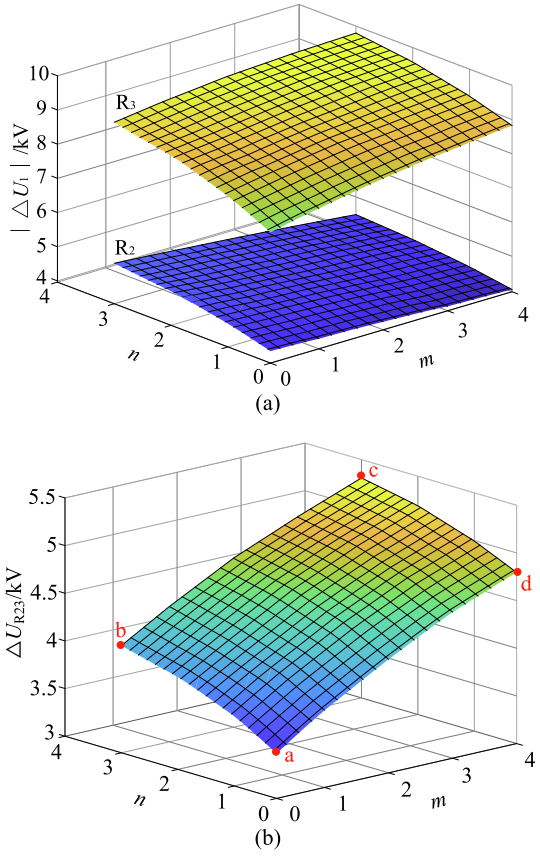


Fig. 7. The effect of DG access on PSFCVs when the fault occurs at the remote-end of L3. (a) PSFCVs of R_3 and R_2 (b) $\triangle U_{R23}$.

In summary, the distribution characteristics of PSFCVs with a high proportion of DG access are as follows:

1) The natural step distribution characteristics of all relays' PSFCVs can directly reflect the fault severity and provide a theoretical basis for realizing fault identification with the unified characteristic equation.

2) The supporting effects provided by DGs change the amplitude of PSFCVs, but its access does not change the distribution characteristics of PSFCVs.

3. Unified characteristic equation based on PSFCV

The combination of the PSFCV's distribution characteristics and the inverse-time characteristic equation allows adaptive coordination of relays without step-by-step setting. Can these functions be realized by using the standard ITOC characteristic equation? Therefore, this section analyzes and gives the functional form and setting principles of the unified inverse-time characteristic equation.

3.1. The standard ITOC characteristic equation and problems with being a unified characteristic equation

1) Characteristic equation: The characteristic equation of traditional ITOC relays is (10) [34].

$$t = \frac{A \times TDS}{(I/I_p)^{\alpha} - 1} \tag{10}$$

where A and α are inverse-time characteristic constants, and TDS is time dial setting. I is the measured current, I_p is the pickup value, and t is tripping time. When $I < I_p$, t is negative and ITOC relay does not act; When $I > I_p$, t is positive and ITOC relay acts. The standard inverse-time characteristic equation is mostly used in distribution network, with $A = 0.14$ and $\alpha = 0.02$.

The fault identification method takes the PSFCVs as the measured value and adopts the standard inverse-time characteristic curve. Analogous to traditional ITOC, the characteristic equation of the proposed method is:

$$t_j = \frac{0.14 \times TDS}{\left(\left| \Delta U'_{1j} \right| / \Delta U_p \right)^{0.02} - 1} \tag{11}$$

where t_j is the tripping time of R_j, $|\triangle U'_{1j}| = k_{dj}|\triangle U_{1j}|$, and $\triangle U_p$ is the pickup value.

In the case of $|\triangle U'_{1j}| > \triangle U_p$, (11) starts and calculates the tripping time. In the case of $|\triangle U'_{1j}| < \triangle U_p$, (11) does not start.

2) Selection principle of $\triangle U_p$ and TDS: The fault identification method without a step-by-step setting adopts a unified $\triangle U_p$ and TDS. The voltage deviation of the three-phase power source of 20 kV and below is ± 7 % of the rated voltage [35]. Considering that the voltage fluctuation caused by load switching and system disturbance may lead to the mis-start of protection, the pickup value $\triangle U_p$ is taken as:

$$\Delta U_p = 1.1 \times 0.07 U_N \tag{12}$$

When the unified $\triangle U_p$ is adopted for the whole network, t_j is a function of TDS and the larger the TDS value, the larger the t_j.

The tripping time interval between adjacent relays should be greater than the minimum coordination time interval (CTI), generally taken as 0.2 ~ 0.5 s [22]. Therefore, the TDS satisfying CTI requirements is:

$$TDS = \max\left\{ \frac{\mathrm{CTI}}{\left(0.14 \times \left(1 \Big/ \left(\Delta U_{j-1}^{'*0.02} - 1 \right) - 1 \Big/ \left(\Delta U_j^{'*0.02} - 1 \right) \right) \right)} \right\} \tag{13}$$

where $\triangle U'^*_j = |\triangle U'_{1j}| / \triangle U_p$.

The relationship between TDS and PSFCVs of adjacent upstream and downstream relays is shown in Fig. 8. (a) shows the relationship between TDS and $|\triangle U'_{1j}|$, $|\triangle U'_{1,\ j\text{-}1}|$. When $|\triangle U'_{1,\ j\text{-}1}|$ is fixed and

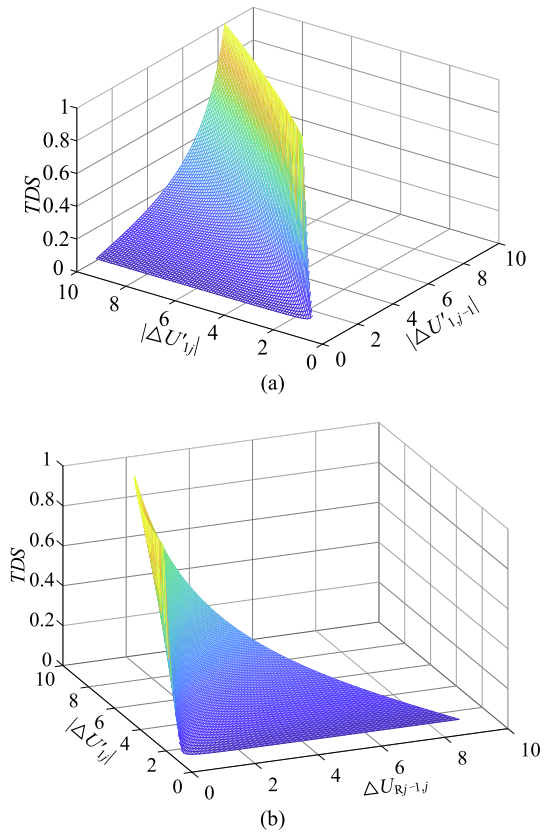


Fig. 8. The relationship between *TDS* and PSFCVs of adjacent upstream and downstream relays. (a) *TDS* and $|\triangle U'_{1j}|$, $|\triangle U'_{1,\,j\text{-}1}|$ (b) *TDS* and $|\triangle U'_{1j}|$, $\triangle U_{Rj\text{-}1,j}$.

$|\triangle U'_{1j}|$ is increased, *TDS* decreases; When $|\triangle U'_{1j}|$ is fixed and $|\triangle U'_{1,\,j\text{-}1}|$ increases, *TDS* increases. Therefore, the *TDS* is related to the PSFCVs amplitude of adjacent relays and the difference between them, as shown in Fig. 8 (b). The value of *TDS* has a nonlinear relationship with the variables $|\triangle U'_{1j}|$ and $\triangle U_{Rj\text{-}1,j}$. It is not that the larger $|\triangle U'_{1j}|$ and $\triangle U_{Rj\text{-}1,j}$, the greater the *TDS* value.

3) Problems with being a unified characteristic equation: The time-PSFCV characteristic curves obtained at points a, b, c and d in Fig. 7 are shown in Fig. 9. TDS_{max3} is not obtained at point a or c where DG output is the minimum or maximum, but at point b where DG_1′s output is 0 and DG_2′s output is maximum. However, this makes it difficult to adopt unified *TDS* across the network. The DG access state and capacity need to be known in advance in order to calculate the maximum *TDS* that satisfies the CTI, which is unrealistic in engineering applications.

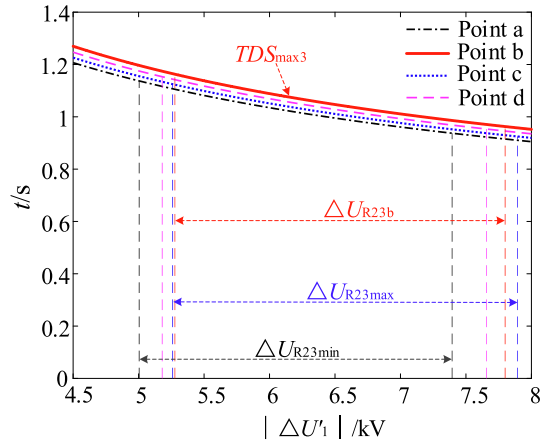


Fig. 9. Time-PSFCV characteristic curves obtained at points a, b, c and d.

3.2. User-defined inverse-time characteristic equation

To achieve a fault sequencing that does not depend on the DG state, the functional form and parameter setting principle of the user-defined inverse-time characteristic equation can only be determined on the basis of the network topology and CTI requirements.

1) Functional form: The PSFCV at the most severe fault can reach U_N, so the range of PSFCV is $[U_N\text{-}\triangle U_p, U_N]$. Let the PSFCV difference between adjacent upstream and downstream relays be $\triangle U_R$, the PSFCVs are:

$$\begin{cases} \left|\Delta U'_{1j}\right|_p = U_N - (p-1)\Delta U_R \\ \left|\Delta U'_{1,\,j-1}\right|_p = U_N - p\Delta U_R \end{cases} \tag{14}$$

where $p = [1, 2, 3, \ldots, P]$, $P = \text{round}((U_N\text{-}\triangle U_p)/\triangle U_R)$.

Bring (14) into (13) to calculate the TDS_j and draw the inverse-time characteristic curve in the $[|\triangle U'_{1,\,j\text{-}1}|_p, |\triangle U'_{1j}|_p]$ by (11). The curves obtained in $[U_N\text{-}P\triangle U_R, U_N]$ are shown in Fig. 10. When U_N is fixed, TDS_j is determined only by $\triangle U_R$ and the larger $\triangle U_R$ is, the smaller the TDS_j is and the smaller t_j is.

However, the curves in $[U_N\text{-}P\triangle U_R, U_N]$ do not satisfy the inverse-time characteristics. Using the relay's requirement for a tripping time, all curves are translated to the specified position to make them satisfy the inverse-time characteristic. Let the minimum tripping time when the relay is used as a back-up protection t_{min}. First, the rightmost curve is moved down to the passing point (U_N, t_{min}), and then the adjacent curves on the left are moved to the passing point ($|\triangle U'_{1,\,j\text{-}1}|_p$, $t_j(p\text{-}1)$- $t_j(p)$ + CTI) in turn to obtain the new curves, as shown in Fig. 11.

From Fig. 11, it is known that t_j is approximated as a straight line in the $[U_N\text{-}\triangle U_p, U_N]$. Therefore, this paper uses the first-order function $y = kx + b$ to obtain the inverse-time equation. k is determined by CTI and $\triangle U_R$, b is determined by k and (U_N, t_{min}). The functional form is:

$$t_j = -\frac{\text{CTI}}{\Delta U_R}\left|\Delta U'_{1j}\right| + \left(t_{min} + \frac{\text{CTI}}{\Delta U_R}U_N\right) \tag{15}$$

When $|\triangle U'_{1j}| > \triangle U_p$, (15) is started and the tripping time is calculated.

$\triangle U_R$ determines the selectivity and rapidity of the tripping time calculated by (15). The selectivity is reflected in the fact that the PSFCV difference of adjacent upstream and downstream relays must be greater than $\triangle U_R$ to ensure that the tripping time difference is greater than CTI. The rapidity is reflected in the fact that $\triangle U_R$ determines the slope of (15), which corresponds to the magnitude of tripping time. The multi-level coordinated relays must ensure selectivity, so the setting principle of minimum $\triangle U_R$ should be given.

2) Setting principle of $\triangle U_R$: Fault sequencing is achieved by the relay of the faulty line and the relays of the adjacent upstream healthy lines with which it is coordinated. The tripping time of the faulty line only needs to be greater than the minimum action time of the backup

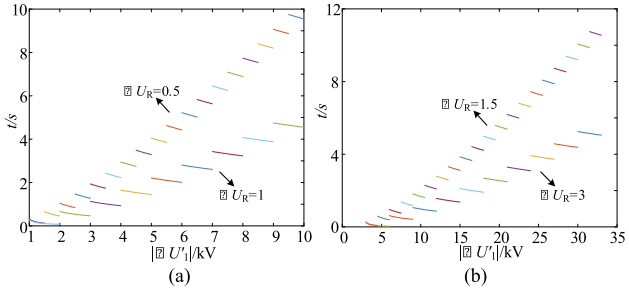


Fig. 10. Curves obtained in $[U_N\text{-}P\triangle U_R, U_N]$. (a) U_N = 10 kV (b) U_N = 33 kV.

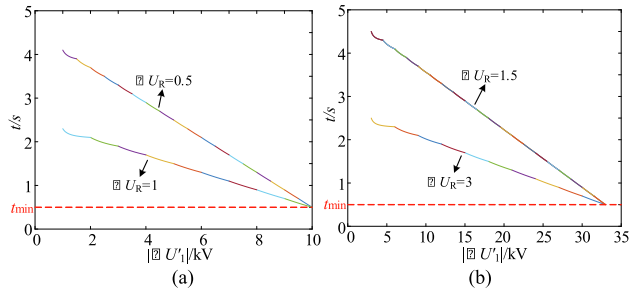


Fig. 11. Curves after translation. (a) U_N = 10 kV (b) U_N = 33 kV.

protection, while the tripping time of the adjacent upstream relays also need to meet the CTI requirements. Therefore, their selection principle for $\triangle U_R$ needs to be discussed separately.

a) Selection principle of $\triangle U_R$ for adjacent upstream healthy lines. As shown in Fig. 4, the fault component currents of the loads and DGs are opposite to the reference direction, resulting in an increase in the PSFCV difference between the adjacent upstream and downstream relays. When no DG is connected on the bus, the PSFCV difference between adjacent upstream and downstream relays is minimal. Therefore, the value of $\triangle U_{R_min}$ set by the network without DG is the lower boundary that satisfies the CTI requirements.

In case of near-end fault, the PSFCV difference between adjacent relays is greater than that of remote-end fault. And the farther the fault point is from the power supply, the smaller the difference is. Therefore $\triangle U_{R_min}$ is obtained by the remote-end fault of the line farthest from the power supply.

When there is no DG access in Fig. 3, $\triangle U_{R_min}$ is obtained from the PSFCV difference when the fault occurs at the remote-end of L3, which is:

$$\Delta U_{R_min} = \min\left(\left|z_1 L_2 \Delta \dot{I}_{12}\right|, \left|z_1 L_1 \Delta \dot{I}_{11}\right|\right) \tag{16}$$

The smaller of L_1 and L_2 in (16) determines $\triangle U_{R_min}$, which takes into account the influence of line length.

When a fault occurs via R_g, $\triangle U_{Rj\text{-}1,j}$ decreases as R_g increases and the t_j calculated by (15) increases. The relay's tripping time should meet the selectivity requirements under any fault condition. To improve the rapidity, the method of obtaining $\triangle U_{R_min}(R_g)$ from different R_g is proposed. Ultimately, the user-defined inverse-time characteristic equations determined by the relationship between $\triangle U_{Rj\text{-}1,j}$ and $\triangle U_{R_min}(R_g)$ is obtained, with the expression as in (17).

$$t_{j.h} = \begin{cases} \left\{ -\dfrac{\mathrm{CTI}}{\Delta U_{R_min}(0)}\left|\Delta U_{1j}^{\cdot}\right| + \left(t_{min} + \dfrac{\mathrm{CTI}}{\Delta U_{R_min}(0)} U_N\right) \right\} \\ \text{where } \Delta U_{Rj\text{-}1,j} > \Delta U_{R_min}(0) \\ \left\{ -\dfrac{\mathrm{CTI}}{\Delta U_{R_min}(R_{Ng})}\left|\Delta U_{1j}^{\cdot}\right| + \left(t_{min} + \dfrac{\mathrm{CTI}}{\Delta U_{R_min}(R_{Ng})} U_N\right) \right\} \\ \text{where } \Delta U_{R_min}(0) > \Delta U_{Rj\text{-}1,j} > \Delta U_{R_min}(R_{Ng}) \\ \vdots \\ \left\{ -\dfrac{\mathrm{CTI}}{\Delta U_{R_min}((h-1)R_{Ng})}\left|\Delta U_{1j}^{\cdot}\right| + \left(t_{min} + \dfrac{\mathrm{CTI}}{\Delta U_{R_min}((h-1)R_{Ng})} U_N\right) \right\} \\ \text{where } \Delta U_{R_min}((h-2)R_{Ng}) > \Delta U_{Rj\text{-}1,j} > \Delta U_{R_min}((h-1)R_{Ng}) \end{cases} \tag{17}$$

where R_{Ng} is the reference value for the change in fault resistance, h is an integer greater than 0.

When $R_{j\text{-}1}$ and $R_{j\text{-}2}$ is used as a backup relay to R_j, three relationships exist between $\triangle U_{Rj\text{-}1,j}$, $\triangle U_{Rj\text{-}2,j\text{-}1}$ and $\triangle U_{R_min}((h\text{-}1)R_g)$. The corresponding curves are used to calculate the tripping time of $R_{j\text{-}1}$ and $R_{j\text{-}2}$, as shown in Fig. 12.

Fig. 12 (c) may lead to a loss of selectivity for $R_{j\text{-}1}$ and $R_{j\text{-}2}$. This paper uses relays to calculate the PSFCV difference with the adjacent downstream 2-level relays to ensure selectivity. The tripping time curve selected by $R_{j\text{-}2}$ is determined by the smaller of $\triangle U'_{Rj\text{-}2,j\text{-}1}$ and $\triangle U'_{Rj\text{-}1,j}$, which are calculated by $R_{j\text{-}2}$. $\triangle U'_{Rj\text{-}2,j\text{-}1}$ is the actual voltage drop of the line between $R_{j\text{-}2}$ and $R_{j\text{-}1}$. $\triangle U'_{Rj\text{-}1,j}$ is generally less than the actual voltage drop of the line between $R_{j\text{-}1}$ and R_j, ensuring absolute selectivity.

b) Selection principle of $\triangle U_R$ for faulty line. The t_j of the faulty line only needs to be greater than t_{min}, and the larger the $\triangle U_R$, the smaller the t_j. Therefore, the expression corresponding to $\triangle U_{R_min}(0)$ is used to calculate the tripping time of the faulty line as:

$$t_{j\text{-}f} = -\frac{\mathrm{CTI}}{\Delta U_{R_min}(0)}\left|\Delta U_{1j}^{\cdot}\right| + \left(t_{min} + \frac{\mathrm{CTI}}{\Delta U_{R_min}(0)} U_N\right) \tag{18}$$

In case of fault, the protection action signal (pas) of the main protection will change from 0 to 1 [36]. Therefore, the pas information is used to determine whether the fault is a faulty line. When pas = 1, the tripping time is calculated by (18); When pas = 0, the tripping time is calculated by (17).

From a) b) it is deduced that the tripping time of each relay is:

$$t_j = \begin{cases} t_{j\text{-}f} & \text{pas} = 1 \\ t_{j\text{-}h} & \text{pas} = 0 \end{cases} \tag{19}$$

The tripping time obtained by (19) ensures that the multilevel coordinated relays meet both selectivity and rapidity requirements, as it consider the influence of line length and fault resistance. Thus, the user-defined inverse-time characteristic equation can realize fault identification without step-by-step setting and adaptive DG access.

4. Fault identification scheme of multilevel relays

The steps of the fault identification scheme are: a) (19) is obtained offline from the network without DGs and it is used as the inverse-time characteristic equation for each ITOC relay. b) For a fault at any point, when pas = 0, $\triangle U_{Rj\text{-}1,j}$ calculated by R_j determines the curve chosen to calculate the tripping time and calculate the tripping time online.

Fig. 13 shows the fault identification for multilevel relays when the same equation is used. The relationship between the PSFCV and the tripping time of relays relative to the fault point f is $|\triangle U'_{13}| > |\triangle U'_{12}| > |\triangle U'_{11}|$, $t_3 > t_2 > t_1$. The closer the relay is to the fault point, the shorter the tripping time. It obtains the action sequencing of the adaptive fault position, using only the natural distribution of the PSFCV. The proposed method protects the entire line length and is applicable to any scenario with or without DG access.

The fault identification scheme essentially an adaptive coordination based on the line voltage drop. Therefore, it is adaptive to the topology with fault point as the source and PSFCVs showing natural difference, such as the multi-terminal power supply system with high penetration DGs (directional elements need to be introduced). Either a unified inverse-time characteristic equation can be used for the whole network or for all relays with a coordinated relationship on a single feeder.

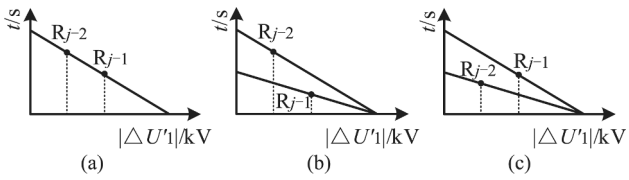


Fig. 12. The curves used to calculate the tripping time of $R_{j\text{-}1}$ and $R_{j\text{-}2}$. (a) $\triangle U_{Rj\text{-}1,j} > \triangle U_{R_min}((h\text{-}1)R_g)$, $\triangle U_{Rj\text{-}2,j\text{-}1} > \triangle U_{R_min}((h\text{-}1)R_g)$ (b) $\triangle U_{Rj\text{-}1,j} > \triangle U_{R_min}((h\text{-}1)R_g)$, $\triangle U_{Rj\text{-}2,j\text{-}1} < \triangle U_{R_min}((h\text{-}1)R_g)$ (c) $\triangle U_{Rj\text{-}1,j} < \triangle U_{R_min}((h\text{-}1)R_g)$, $\triangle U_{Rj\text{-}2,j\text{-}1} > \triangle U_{R_min}((h\text{-}1)R_g)$.

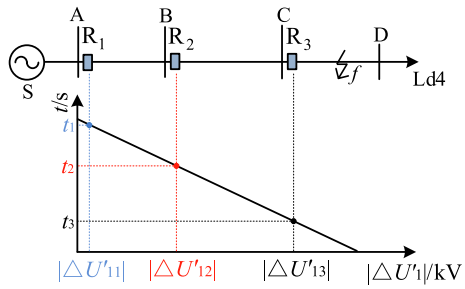


Fig. 13. Fault identification for multilevel relays.

5. Simulation results

The typical radial distribution network and the distribution system of modified IEEE 30-bus test system is analyzed [8,37]. They are established in PSCAD/EMTDC, as shown in Fig. 14.

The typical radial network in Fig. 14 (a) has two feeders. The ends of the two feeders are connected through sectionalizing switch S_1, which is open during normal operation. Network parameters are shown in Table 1. S_{DG_base} is the rated power of the individual DG and I_N is the rated current. $m_1 \sim m_3$ denotes the number of $DG_1 \sim DG_3$.

The network parameters in Fig. 14 (b) are available in [37]. The line length from L1 to L9 is 100 km and from L9 to L21 is 50 km. B1, B8 and B15 are connected to G_1, G_2 and G_3 respectively via 132/33 kV transformers. The system is equipped with 42 inverse-time relays. Different types of DGs with high penetration are connected to the grid, and their distribution is shown in Table 2.

The sampling frequency is 2 kHZ, CTI = 0.2 s, t_{min} = 0.5 s.

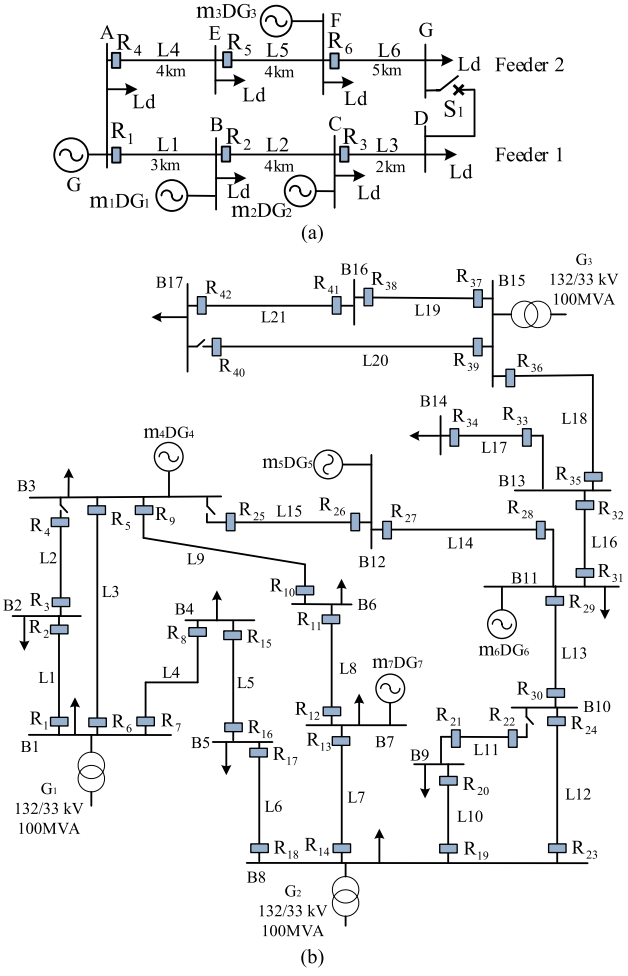


Fig. 14. Simulation models. (a) Typical radial distribution network (b) Distribution system of modified IEEE 30-bus test system.

Table 1
Network parameters in Fig. 14 (a)

Network parameters	
U_N	10 kV
Line model	centralized parameter model
z_1	0.27 + j0.352 Ω/km
Ld	1 + j0.1 MVA
$DG_1 \sim DG_3$	PV with LVRT capability
S_{DG_base}	1 MVA
I_{max}	$1.1I_N$

5.1. Typical radial network

1) Unified inverse-time characteristic equation for the whole network: There are 6 relays in the network, so the whole network uses a unified inverse-time characteristic equation for fault sequencing. Too many relays with coordination relationship on a feeder participate in fault sequencing, resulting in a reduction in $\triangle U_{R_min}(R_g)$ and an increase in tripping time. Therefore, only 3 relays with a coordinated relationship on each feeder are sequenced in this paper, and the remote-end faults of the downstream relays are used to calculate $\triangle U_{R_min}$.

System reconfiguration includes load transfer, load recovery after a fault or maintenance [10]. The opening and closing of S_1 changes the network topology, e.g. when S_1 is closed and the circuit breaker B1 at R_1 is open, R_3 becomes the adjacent upperstream of R_2. Therefore, the setting of $\triangle U_{R_min}$ should take into account the state of S_1. When the state of S_1 is different, the fault positions of a feeder seeking its corresponding $\triangle U_{R_min}$ are different, as shown in Fig. 15. $f1 \sim f8$ are the remote faults of the corresponding lines.

The network-wide minimum $\triangle U_{R_min}$ is determined by the fault position corresponding to the minimum value among all fault points for the same fault condition, and this fault position is recorded as $f_{\text{-end}}$. The $\triangle U_{R_min}$ on a feeder and the $f_{\text{-end}}$ are shown in Table 3.

Taking $R_{Ng} = 5\ \Omega$ and $h = 2$. When the fault occurs at $f8$, the unified inverse-time characteristic equation is:

$$t_{j_f} = -0.23\left|\Delta U'_{1j}\right| + 2.7$$

$$t_{j_h} = \begin{cases} -0.23\left|\Delta U'_{1j}\right| + 2.7 & \Delta U_{Rj\text{-}1,\,j} > 0.91 \\ -0.34\left|\Delta U'_{1j}\right| + 3.89 & 0.59 < \Delta U_{Rj\text{-}1,\,j} < 0.91 \\ -0.48\left|\Delta U'_{1j}\right| + 5.25 & 0.42 < \Delta U_{Rj\text{-}1,\,j} < 0.59 \end{cases} \quad (20)$$

2) Adaptability analysis of DG access and system reconfiguration: Fig. 16 shows the fault component current waveform of phase a of R_2, R_3 and DG_2 when the voltage sag is 100 %. When the influence of the load current is ignored, the sum of the fault component currents of R_2 and DG_2 in the fault steady state is equal to R_3, which verifies that the fault component current of DG_2 flows into the system.

where $\triangle i_2$, $\triangle i_3$, and $\triangle i_{g2}$ represent the time domain current flowing through R_2, R_3, and DG_2, respectively.

The adaptability of the proposed method to DG access is verified by

Table 2
DGs' distribution in Fig. 14 (b)

DG	Type	Connection Bus	S_{DG_base}
DG_4	PV	B3	4
DG_5	PMSG	B12	1.5
DG_6	PV	B11	2
DG_7	PMSG	B7	2

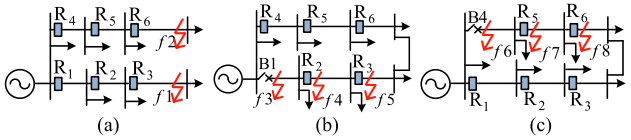


Fig. 15. Fault positions of the feeder in different S_1 states. (a) S_1 is open (b) S_1 is closed and B_1 is open (c) S_1 is closed and B_4 is open.

Table 3
The $\triangle U_{R_min}$ on a feeder and the $f_{\text{-end}}$

Fig. 15	Fault position	PSFCV differences/kV	$\triangle U_{R_min}$	$f_{\text{-end}}$
(a)	$f1$	$\lvert\triangle U'_{R23}\rvert$ =2.04, $\lvert\triangle U'_{R12}\rvert$ =1.43	1.43	$f8$
	$f2$	$\lvert\triangle U'_{R56}\rvert$ =1.59, $\lvert\triangle U'_{R45}\rvert$ =1.47	1.47	
(b)	$f3$	$\lvert\triangle U'_{R32}\rvert$ =1.05, $\lvert\triangle U'_{R63}\rvert$ =1.65	1.05	
	$f4$	$\lvert\triangle U'_{R63}\rvert$ =1.89, $\lvert\triangle U'_{R56}\rvert$ =1.000	1.000	
	$f5$	$\lvert\triangle U'_{R56}\rvert$ =1.24, $\lvert\triangle U'_{R45}\rvert$ =1.13	1.13	
(c)	$f6$	$\lvert\triangle U'_{R65}\rvert$ =1.06, $\lvert\triangle U'_{R36}\rvert$ =1.73	1.06	
	$f7$	$\lvert\triangle U'_{R36}\rvert$ =2.07, $\lvert\triangle U'_{R23}\rvert$ =1.07	1.07	
	$f8$	$\lvert\triangle U'_{R23}\rvert$ =1.32, $\lvert\triangle U'_{R12}\rvert$ =0.91	0.91	

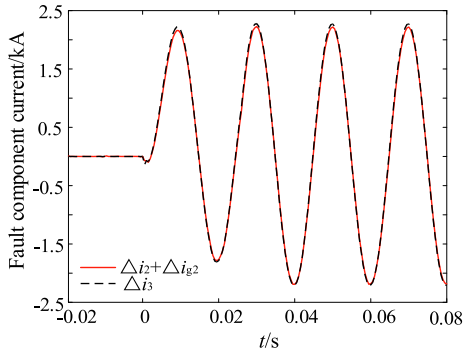


Fig. 16. The fault component current waveform of phase a.

varying the DG penetration. Table 4 gives the tripping time of $R_1 \sim R_3$ on Feeder 1 for different fault positions and different DG penetration when S_1 is open. Table 5 gives the tripping times of R_2, R_3 and R_6 when S_1 is closed and B_1 is open. The fault is a LLL metallic short-circuit fault.

The tripping time of relays calculated by (20) meets the requirements of CTI. Fig. 17 shows the value of $\triangle U_{Rj\text{-}1,j}$ for different DG penetrations and when the fault occurs at $f1$ and $f3$. The PSFCV difference of adjacent upstream and downstream relays under different DG penetrations is greater than when there is no DG access, verifying that the proposed method is adaptive to DG access.

Comparing Table 4 and Table 5, it can be seen that the system reconfiguration makes the coordination relationship between R_2 and R_3 reversed. The existing adaptive relays must change their settings in response to changes in circuit load patterns and conditions. After system reconfiguration, the PSFCV distribution characteristic still has the property that the closer to the fault point, the larger the value. The tripping time automatically reflect the upstream and downstream correlation of all relays. Therefore, the scheme is able to adapt to network topology changes without switching settings.

3) Simulation verification of different fault conditions: Table 6 shows the fault identification results of R_5, R_6 and R_3 under different fault types and fault resistances when the fault occurs at $f6$. The penetration of DGs is $m_1 = 0$, $m_2 = 4$, $m_3 = 2$.

Table 6 verifies the correctness of the fault identification scheme for different fault conditions. The difference in tripping time for each relay under different fault types is significantly reduced after the k_{dj} correction. R_{Ng} and h determine the maximum fault resistance of the proposed method in the whole network,which is hR_{Ng}. The minimum difference of PSFCV corresponding to hR_{Ng} obtained from (17) is $0.2/\triangle U_{R_min}(hR_{Ng})$.

When $R_g = 10\ \Omega$ and LLG fault occurs, $\triangle U_{R65} = 0.75$ kV and the tripping time of R_6 and R_3 would be calculated by $t_j = -0.34|\triangle U'_{1j}| + 3.89$. Therefore, the fault resistance of a feeder is greater than hR_{Ng}.

5.2. Modified IEEE 30-bus test system

1) Unified inverse-time characteristic equation for each feeder: The modified IEEE 30-bus test system is a multi-terminal power supply system, and directional elements have been considered in the coordination of relays. As the network contains 42 relays, a unified inverse-time characteristic equation for the whole network would affect the rapidity of fault identification. Therefore, an inverse-time equation is derived for each feeder. Starting with G_1, G_2 and G_3 respectively, the relays with coordination on each feeder are shown in Fig. 18. $f9$ to $f20$ are remote-end faults.

Each dashed box gives the relay's numbers with 3 or more levels of coordination and the fault positions for the calculation $\triangle U_{R_min}$. However, both dashed boxes b_5 and b_6 contain R_{28} and R_{33}. To realize fault identification of adaptive fault position, all relays in b_5 and b_6 should adopt a unified equation.

Taking $R_{Ng} = 5\ \Omega$ and $h = 2$. The inverse-time characteristic equations of dashed boxes $b_1 \sim b_6$ are (21)~(25) respectively.

Table 4
Tripping time of $R_1 \sim R_3$ on feeder 1 for different fault positions and different DG penetration when S_1 is open

DG penetration	Faulty line	Fault position	PSFCVs/kV			Tripping time/s		
			$\lvert\triangle U'_{13}\rvert$	$\lvert\triangle U'_{12}\rvert$	$\lvert\triangle U'_{11}\rvert$	t_3	t_2	t_1
$m_1 = 0$	L3	Near-end	8.35	5.74	3.91	0.78	1.38	1.80
$m_2 = 4$		Midpoint	7.655	5.255	3.584	0.94	1.49	1.88
$m_3 = 0$		Remote-end	7.09	4.86	3.31	1.07	1.58	1.94
	L2	Near-end	–	8.18	5.58	–	0.82	1.42
		Midpoint	–	6.74	4.60	–	1.15	1.64
		Remote-end	–	5.74	3.91	–	1.38	1.80
$m_1 = 2$	L3	Near-end	8.24	5.60	3.80	0.80	1.41	1.83
$m_2 = 2$		Midpoint	7.56	5.11	3.48	0.96	1.52	1.90
$m_3 = 0$		Remote-end	6.99	4.75	3.22	1.09	1.61	1.96
	L2	Near-end	–	8.22	5.60	–	0.81	1.41
		Midpoint	–	6.67	4.55	–	1.17	1.65
		Remote-end	–	5.56	3.79	–	1.42	1.83
$m_1 = 2$	L3	Near-end	8.49	5.78	3.93	0.75	1.37	1.80
$m_2 = 4$		Midpoint	7.76	5.28	3.60	0.92	1.49	1.87
$m_3 = 0$		Remote-end	7.12	4.82	3.28	1.06	1.59	1.95
	L2	Near-end	–	8.34	5.69	–	0.78	1.39
		Midpoint	–	6.81	4.64	–	1.13	1.63
		Remote-end	–	5.80	3.95	–	1.37	1.79

Table 5
Tripping time of R_3, R_6 and R_5 when S_1 is closed and B_1 is open

DG penetration	Faulty line	Fault position	PSFCVs/kV			Tripping time/s		
			$\lvert\triangle U'_{12}\rvert$	$\lvert\triangle U'_{13}\rvert$	$\lvert\triangle U'_{16}\rvert$	t_2	t_3	t_6
$m_1 = 0$	L1	Near-end	7.86	6.21	3.99	0.89	1.27	1.78
$m_2 = 4$		Midpoint	7.25	5.71	3.67	1.03	1.39	1.86
$m_3 = 0$		Remote-end	6.70	5.22	3.37	1.16	1.50	1.92
	L2	Near-end	–	7.95	5.12	–	0.87	1.52
		Midpoint	–	6.99	4.50	–	1.09	1.67
		Remote-end	–	6.21	3.99	–	1.27	1.78
$m_1 = 0$	L1	Near-end	7.29	5.81	3.49	1.02	1.36	1.90
$m_2 = 0$		Midpoint	6.75	5.39	3.25	1.15	1.46	1.95
$m_3 = 4$		Remote-end	6.28	5.01	2.99	1.26	1.55	2.01
	L2	Near-end	–	7.37	4.52	–	1.01	1.66
		Midpoint	–	6.50	3.90	–	1.20	1.80
		Remote-end	–	5.81	3.49	–	1.36	1.90
$m_1 = 0$	L1	Near-end	8.26	6.39	3.89	0.80	1.23	1.81
$m_2 = 4$		Midpoint	7.57	5.81	3.54	0.96	1.36	1.89
$m_3 = 4$		Remote-end	6.97	5.34	3.25	1.10	1.47	1.95
	L2	Near-end	–	8.35	5.14	–	0.78	1.52
		Midpoint	–	7.28	4.44	–	1.03	1.68
		Remote-end	–	6.39	3.89	–	1.23	1.81

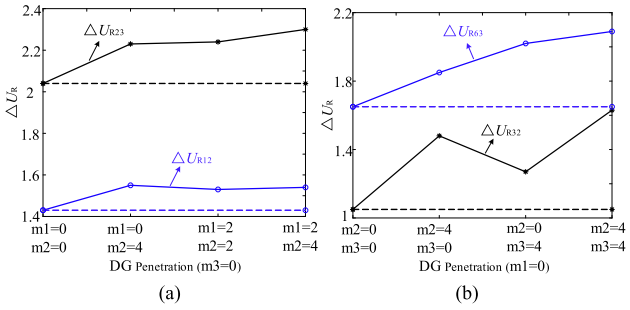


Fig. 17. The value of $\triangle U_{Rj\text{-}1,j}$ for different DG penetrations. (a) When the fault occurs at *f*1 (b) When the fault occurs at *f*3.

Table 6
Fault identification results of R_5, R_6 and R_3 under different fault types and fault resistances

R_g/Ω	Fault type	PSFCVs/kV			Tripping time/s		
		$\lvert\triangle U'_{15}\rvert$	$\lvert\triangle U'_{16}\rvert$	$\lvert\triangle U'_{13}\rvert$	t_5	t_6	t_3
0	LLL	6.42	4.98	2.92	1.22	1.56	2.03
	LL	6.47	5.07	3.10	1.21	1.53	1.99
	LLG	6.48	5.10	3.05	1.21	1.53	2.00
5	LLL	5.75	4.47	2.64	1.38	1.67	2.09
	LL	5.46	4.28	2.57	1.44	1.72	2.11
	LLG	4.66	3.66	2.21	1.63	1.86	2.19
10	LLL	5.15	3.99	2.37	1.52	1.78	2.16
	LL	4.67	3.66	2.23	1.63	1.86	2.19
	LLG	3.56	2.81	1.70	1.88	2.93	3.31

$$b_1:\begin{cases} t_{j_f} = -0.07\left|\Delta U'_{1j}\right| + 2.7 \\ t_{j_h} = \begin{cases} -0.07\left|\Delta U'_{1j}\right| + 2.7 \ \Delta U_{Rj\text{-}1,j} > 3.01 \\ -0.16\left|\Delta U'_{1j}\right| + 5.54 \ 1.30 < \Delta U_{Rj\text{-}1,j} < 3.01 \\ -0.27\left|\Delta U'_{1j}\right| + 9.39 \ 0.74 < \Delta U_{Rj\text{-}1,j} < 1.30 \end{cases} \end{cases} \quad (21)$$

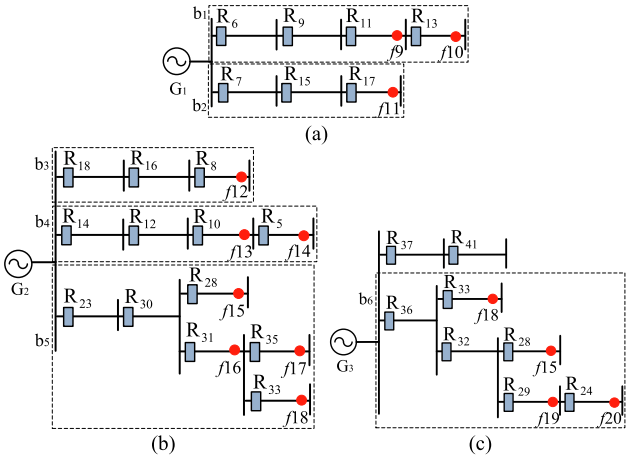


Fig. 18. The relays with coordination on each feeder starting with G_1,G_2 and G_3. (a) G_1 (b) G_2 (c) G_3.

$$b_2:\begin{cases} t_{j_f} = -0.04\left|\Delta U'_{1j}\right| + 1.6 \\ t_{j_h} = \begin{cases} -0.04\left|\Delta U'_{1j}\right| + 1.6 \ \Delta U_{Rj\text{-}1,j} > 6.01 \\ -0.08\left|\Delta U'_{1j}\right| + 2.91 \ 2.74 < \Delta U_{Rj\text{-}1,j} < 6.01 \\ -0.13\left|\Delta U'_{1j}\right| + 4.72 \ 1.56 < \Delta U_{Rj\text{-}1,j} < 2.74 \end{cases} \end{cases} \quad (22)$$

$$b_3:\begin{cases} t_{j_f} = -0.07\left|\Delta U'_{1j}\right| + 2.58 \\ t_{j_h} = \begin{cases} -0.07\left|\Delta U'_{1j}\right| + 2.58 \ \Delta U_{Rj\text{-}1,j} > 3.17 \\ -0.12\left|\Delta U'_{1j}\right| + 4.42 \ 1.68 < \Delta U_{Rj\text{-}1,j} < 3.17 \\ -0.19\left|\Delta U'_{1j}\right| + 6.47 \ 1.10 < \Delta U_{Rj\text{-}1,j} < 1.68 \end{cases} \end{cases} \quad (23)$$

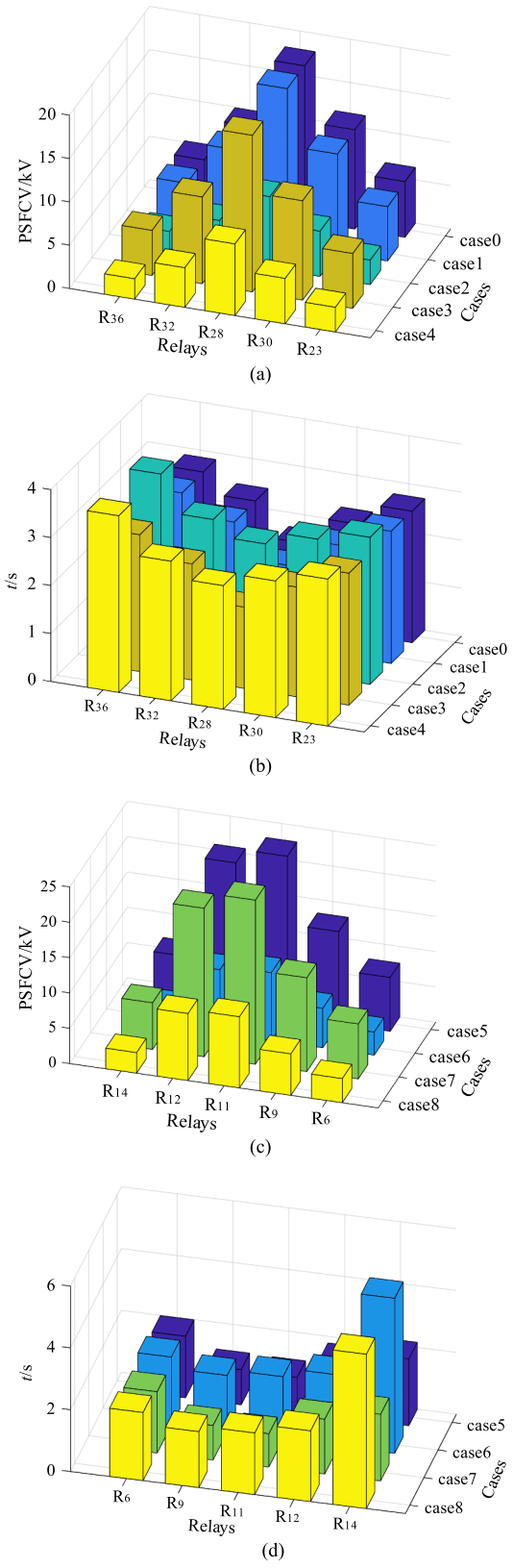


Fig. 19. Fault identification results for different fault conditions. (a) PSFCVs of relays when the fault occurs at the midpoint of L14 (b) Tripping time of relays when the fault occurs at the midpoint of L14 (c) PSFCVs of relays when the fault occurs at the midpoint of L8 (d) Tripping time of relays when the fault occurs at the midpoint of L8.

Table 7
DG penetration, fault resistance, fault position and fault type for the cases

Cases	R_g/Ω	DG penetration	Fault position	Fault type
Case 0	0	0	Midpoint of L4	LLL
Case 1	0	$m_4 = 0$, $m_5 = 20$		
Case 2	5	$m_6 = 2$, $m_7 = 10$		
Case 3	0	$m_4 = 4$, $m_5 = 20$		
Case 4	5	$m_6 = 0$, $m_7 = 5$		
Case 5	0	$m_4 = 2$, $m_5 = 8$	Midpoint of L8	LL
Case 6	10	$m_6 = 2$, $m_7 = 10$		
Case 7	0	$m_4 = 4$, $m_5 = 20$		
Case 8	10	$m_6 = 2$, $m_7 = 20$		

$$b_4: \begin{cases} t_{j_f} = -0.06\left|\Delta U'_{1j}\right| + 2.38 \\ t_{j_h} = \begin{cases} -0.06\left|\Delta U'_{1j}\right| + 2.38 & \Delta U_{Rj\text{-}1,j} > 3.50 \\ -0.12\left|\Delta U'_{1j}\right| + 4.18 & 1.79 < \Delta U_{Rj\text{-}1,j} < 3.50 \\ -0.17\left|\Delta U'_{1j}\right| + 6.07 & 1.18 < \Delta U_{Rj\text{-}1,j} < 1.79 \end{cases} \end{cases} \tag{24}$$

$$b_5, b_6: \begin{cases} t_{j_f} = -0.09\left|\Delta U'_{1j}\right| + 3.32 \\ t_{j_h} = \begin{cases} -0.09\left|\Delta U'_{1j}\right| + 3.32 & \Delta U_{Rj\text{-}1,j} > 2.34 \\ -0.11\left|\Delta U'_{1j}\right| + 3.94 & 1.91 < \Delta U_{Rj\text{-}1,j} < 2.34 \\ -0.13\left|\Delta U'_{1j}\right| + 4.83 & 1.52 < \Delta U_{Rj\text{-}1,j} < 1.91 \end{cases} \end{cases} \tag{25}$$

2) Adaptability analysis of fault identification: The PSFCVs and fault identification results of relays for different fault conditions are given in Fig. 19. The DG penetration, fault resistance, fault position and fault type for the cases are shown in Table 7.

When the fault occurs at the midpoint of L14, there are 2 feeders on which the PSFCV of the relays has a natural distribution characteristic. Fig. 19 (a) and (b) places R_{28} in the middle which is the closest to the fault point, where it has a coordinated relationship with R_{32}, R_{36} on the left and R_{30}, R_{23} on the right, respectively. When the fault occurs at the midpoint of L8, R_{11}, R_9 and R_6 are coordinated, as are R_{12} and R_{14}. The simulation verifies that the multilevel coordinated relays achieve adaptive coordination under any fault condition. Therefore, the proposed fault identification scheme can meet the CTI requirements under different fault conditions and adapts to the DG access.

6. Conclusions

This paper analyzes the distribution characteristics of PSFCVs in radial networks with DG access, and a fault identification method with a unified inverse-time characteristic equation is proposed. Using a step-by-step procedure for settings is unnecessary, and all relays adapt to the fault severity according to the calculated tripping time. Based on theoretical analysis and simulation verification, the following conclusions are drawn:

1) By improving the setting process of traditional ITOC relay, a unified characteristic equation expression for all relays can be obtained, which is easy to implement.
2) Relying on only the natural distribution of the PSFCVs, the adaptive coordination of all relays is realized without communication.
3) Compared to traditional methods, the distribution characteristics of PSFCV are not affected by DG access, and the proposed fault identification method can adapt to the high proportion of DG access.

4) The proposed method is self-adaptive to topology changes and is adaptive to the network topology of PSFCVs meeting the natural distribution characteristics.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

[1] Hussain B, Sharkh SM, Hussain S, Abusara MA. An adaptive relaying scheme for fuse saving in distribution networks with distributed generation. IEEE Trans Power Del 2013;28(2):669–77.
[2] Noghabi AS, Mashhadi HR, Sadeh J. Optimal coordination of directional overcurrent relays considering different network topologies using interval linear programming. IEEE Trans Power Del 2010;25(3):1348–54.
[3] Li B, Li C, Guo F, Xin Y. Overcurrent protection coordination in a power distribution network with the active superconductive fault current limiter. IEEE Transactions on Appl Supercond 2014;24(5):1–4.
[4] Zarei SF, Parniani M. A comprehensive digital protection scheme for low-voltage microgrids with inverter-based and conventional distributed generations. IEEE Trans Power Del 2017;32(1):441–52.
[5] Noghabi AS, Sadeh J, Mashhadi HR. Considering different network topologies in optimal overcurrent relay coordination using a hybrid GA. IEEE Trans Power Del 2009;24(4):1857–63.
[6] Azizi S, Liu G, Dobakhshari AS, Terzija V. Wide-area backup protection against asymmetrical faults using available phasor measurements. IEEE Trans Power Del 2020;35(4):2032–9.
[7] Zare J, Aminifar F, Sanaye-Pasand M. Communication-constrained regionalization of power systems for synchrophasor-based wide-area backup protection scheme. IEEE Trans Smart Grid 2015;6(3):1530–8.
[8] Yu K, Lin X, Li H, Zhao H, Li Z, Huang J, et al. A voltage correctio based inverse-time overcurrent protection scheme considering distributed generation stable infeed effect. Proc CSEE 2018;38(3):716–26.
[9] Solati Alkaran D, Vatani MR, Sanjari MJ, Gharehpetian GB, Naderi MS. Optimal overcurrent relay coordination in interconnected networks by using fuzzy-based GA Method. IEEE Trans Smart Grid 2018;9(4):3091–101.
[10] Friend F, et al. Effect of distribution automation on protective relaying. 2014 67th annual conference for protective relay engineers 2014;193–228.
[11] Yousaf M, Jalilian A, Muttaqi KM, Sutanto D. An adaptive overcurrent protection scheme for dual-setting directional recloser and fuse coordination in unbalanced distribution networks with distributed generation. IEEE Trans Ind Appl 2022;58 (2):1831–42.
[12] Seegers T, et al. Impact of distributed resources on distribution relay protection. A report to the line protection subcommittee of the power system relay committee of the IEEE power engineering society 2004:1–26.
[13] Coffele F, Booth C, Dyśko A. An Adaptive Overcurrent Protection Scheme for Distribution Networks. IEEE Trans Power Del 2015;30(2):561–8.
[14] Momesso AE, Bernardes WMS, Asada EN. Fuzzy adaptive setting for time-current-voltage based overcurrent relays in distribution systems. Int J Elect Power Energy Syst 2019;108:135–44.
[15] Jamali S, Borhani-Bahabadi H. Non-communication protection method for meshed and radial distribution networks with synchronousbased DG. Int J Elect Power Energy Syst 2017;93:468–78.
[16] Huang W, Tai N, Zheng X, Fan C, Yang X, Kirby BJ. An impedance protection scheme for feeders of active distribution networks. IEEE Trans Power Del 2014;29 (4):1591–602.
[17] Rezaei N, Uddin MN, Amin IK, Othman ML, Marsadek M. Genetic algorithm-based optimization of overcurrent relay coordination for improved protection of DFIG operated wind farms. IEEE Trans Ind Appl 2019;55(6):5727–36.
[18] Alam MN, Das B, Pant V. A comparative study of metaheuristic optimization approaches for directional overcurrent relays coordination. Elect Power Syst Res 2015;128:39–52.
[19] Ghotbi-Maleki M, Chabanloo RM, Zeineldin HH, Hosseini Miangafsheh SM. Design of setting group-based overcurrent protection scheme for active distribution networks using MILP. IEEE Trans Smart Grid 2021;12(2):1185–93.
[20] Brunelloet G, et al. Summary of C37.230-2007, IEEE guide for protective relay applications to distribution lines. By line protection subcommittee working group D5 Of the IEEE power engineering society - power system relaying committee 2008:1–11.
[21] Alam MN. Adaptive protection coordination scheme using numerical directional overcurrent relays. IEEE Trans Ind Inform 2019;15(1):64–73.
[22] Kiliçkiran HC, Şengör İ, Akdemir H, Kekezoğlu B, Erdinç O, Paterakis NG. Power system protection with digital overcurrent relays: A review of non-standard characteristics. Electr Power Syst Res 2018;164:89–102.
[23] Sharaf M, Zeineldin HH, Ibrahim DK, EL-Zahab EEL-DA. A proposed coordination strategy for meshed distribution systems with DG considering user-defined characteristics of directional inverse time overcurrent relays. Elect Power Energy Syst 2015;65:49–58.
[24] Keil T, Jager J. Advanced coordination method for overcurrent protection relays using nonstandard tripping characteristics. IEEE Trans Power Del 2008;23(1):52–7.
[25] He Z, Zhang Z, Chen W, Malik OP, Yin X. Wide-area backup protection algorithm based on fault component voltage distribution. IEEE Trans Power Del 2011;26(4): 2752–60.
[26] Gao H, Li J, Xu B. Principle and implementation of current differential protection in distribution networks with high penetration of DGs. IEEE Trans Power Del 2017; 32(1):565–74.
[27] aramasu VY, Wu B, Alepuz S, Kouro S. Predictive control for low-voltage ride-through enhancement of three-level-boost and NPC-converter-based PMSG wind turbine. IEEE Trans Ind Electron 2014;61(12):6832–43.
[28] Bae Y, Vu T-K, Kim R-Y. Implemental Control Strategy for Grid Stabilization of Grid-Connected PV System Based on German Grid Code in Symmetrical Low-to-Medium Voltage Network. IEEE Trans Energy Convers 2013;28(3):619–31.
[29] Timbus AV, Rodriguez P, Teodorescu R, Liserre M, Blaabjerg F. Control strategies for distributed power generation systems operating on faulty grid. IEEE Int Symp Ind Electron 2006:1601–7.
[30] Zhang F, Mu L. A Fault detection method of microgrids with grid-connected inverter interfaced distributed generators based on the PQ control strategy. IEEE Trans Smart Grid 2019;10(5):4816–26.
[31] Ge Y. Principle and technology of new relay protection and fault location. Xi'an: Xi'an Jiaotong University Press; 2007.
[32] Caire R, Retiere N, Martino S, Andrieu C, Hadjsaid N. Impact assessment of LV distributed generation on MV distribution network. Proc IEEE PES Summer Meeting 2002;3:1423–8.
[33] Wang Z, Zhu S, Zhou S, Huang R, Wang L. Impact of distributed generation on distribution system voltage profile. Automation of Electric Power Systems 2004;28 (16):56–60.
[34] Ojaghi M, Ghahremani R. Piece-wise linear characteristic for coordinating numerical overcurrent relays. IEEE Trans Power Del 2017;32(1):145–51.
[35] "Power quality-deviation of supply voltage," GB/T 12325-2008.
[36] "Standardization design specification for 10 kV ~110 (66) kV transmission line protection and auxiliary equipments," QGDW 10766-2015.
[37] Univ. Washington. (2006, Mar.). Power Systems Test Case Archive [Online]. Available: http://www.ee.washington.edu/research/.